ECE 431: Optoelectronics

Semiconductor Laser Simulation Assignment

Electronics and Electrical Communications Dept.

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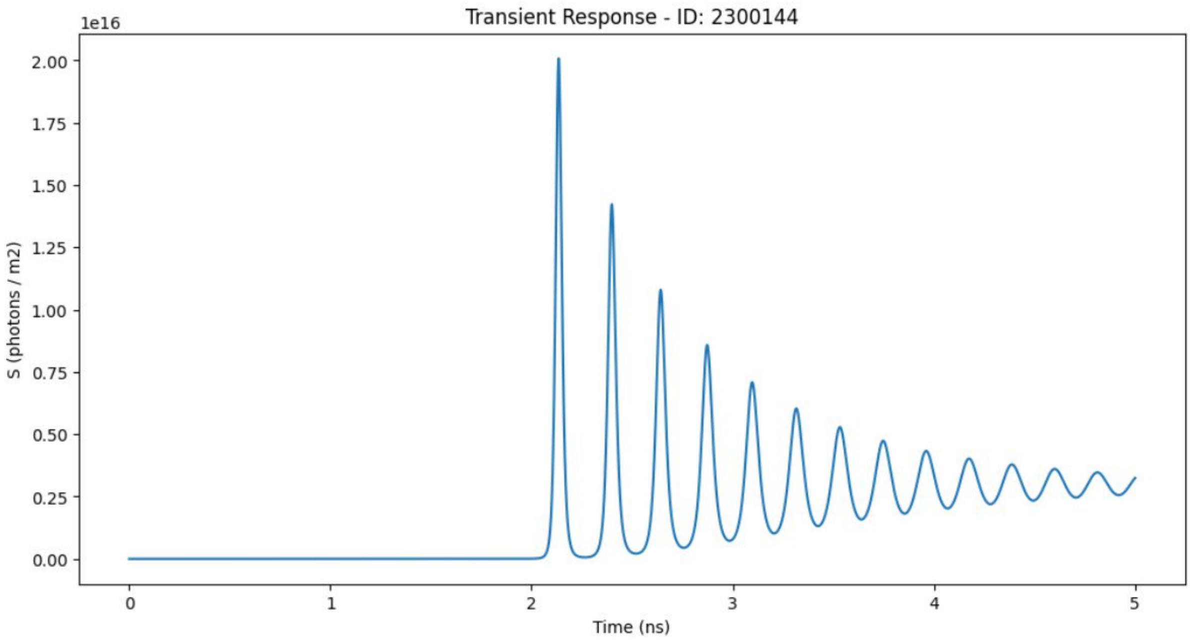
Task A: Transient Response

The transient response of the laser system was simulated to observe the behavior when the injection current I is stepped from 0 to 50mA at t = 1 ns. The numerical solution to the rate equations demonstrates the photon density S(t) over a duration of 5ns.

- **Observation:**

The generated plot clearly shows the relaxation oscillations (ringing) associated with the turn-on event before reaching steady state.

- **Plot the Photon Density S(t) over a duration of 5*ns*:**



Task B: L-I Static Characteristic

- **Simulation Description:**

To analyze the steady-state behavior and derive the Light-Current (L-I) characteristic curve, a parameter sweep of the injection current I was performed from 0 to 60 mA. The numerical solution was evaluated at steady-state to compute the single-facet output power.

$$S = \frac{\tau_{ph}}{ed}(J - J_{th}) \quad (1)$$

$$P = \eta(I - I_{th}) \quad (2)$$

$$\eta = \frac{1}{2L} ln \frac{1}{R_1} \frac{h\nu v_g \tau_{ph}}{e} \quad (3)$$

From (1), (2) and (3):

$$P = \frac{1}{2} ln \frac{1}{R1} h\nu v_g dwS \quad (4)$$

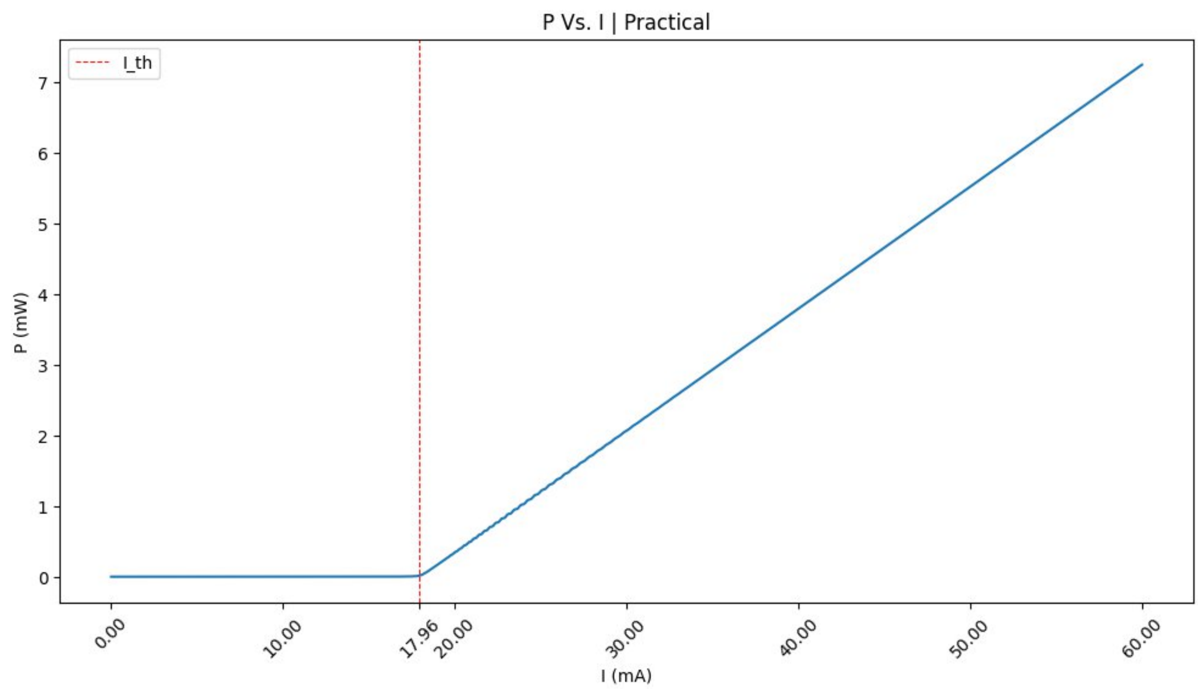
- **Steady-State Observation:**

The steady-state values of power are recorded to be used in P Vs. I plots next.

- **Observation:**

The simulated P-I curve (Practical) demonstrates a clear threshold knee, indicating the transition from spontaneous to stimulated emission.

- **Plot the Output Power P_{out} vs. Current I (practical):**



- **Simulated Threshold Current:**

From the practical L-I curve, the threshold current is numerically observed at the knee of the curve (marked as I_{th} on the plot).

- **Theoretical Hand Analysis:**

To verify the simulation, we calculate the theoretical threshold current (I_{th}) using the cavity parameters derived from Student ID (2300144).

- **Total loss ($\boldsymbol{\alpha r}$)**

$$\alpha r = \alpha int + \frac{1}{L} ln\left(\frac{1}{R}\right) = 40 + \frac{1}{0.032} ln\left(\frac{1}{0.38}\right) = 70.24\ cm^{-1}$$

- **Threshold Carrier Density ($\boldsymbol{N_{th}}$)**

$$N_{th} = N_{tr} + \frac{\alpha r}{\Gamma B} = 10^{18} + \frac{70.24}{0.3 \times 2.5 \times 10^{-16}} = 1.936 \times 10^{18} cm^{-3}$$

- **Theoretical Threshold Current ($\boldsymbol{I_{th}}$)**

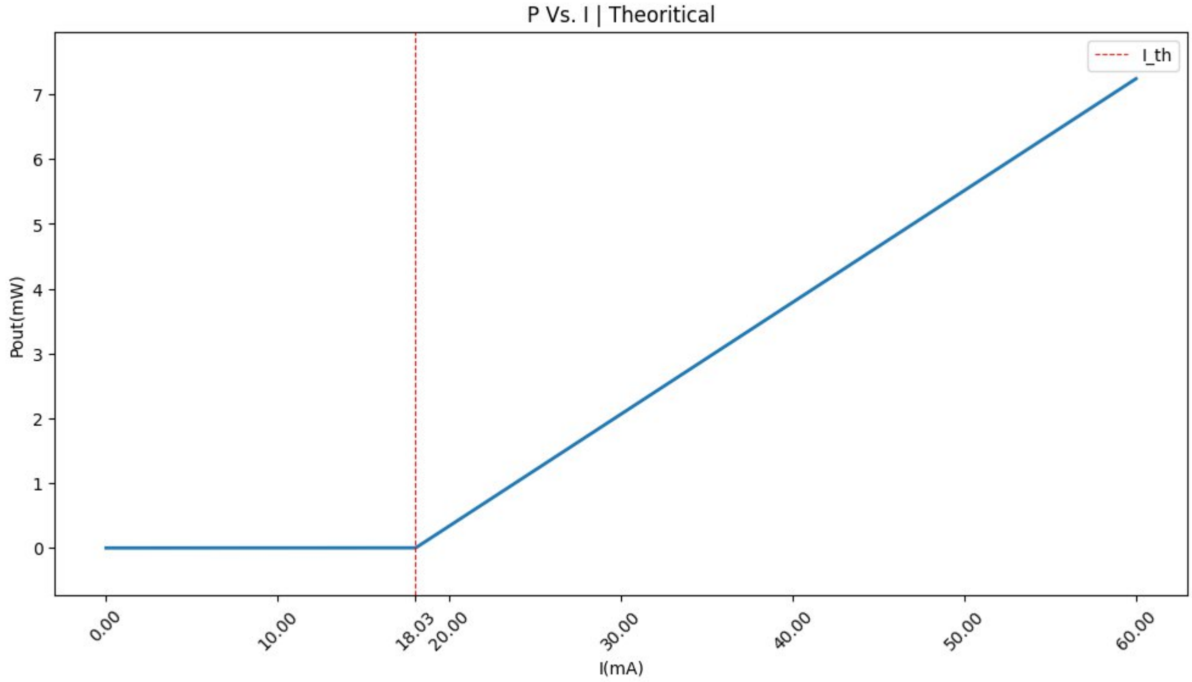
$$I_{th} = \frac{q * V * N_{th}}{\tau} = \frac{q * \left(2 * 10^{-4}\right) * \left(0.2 * 10^{-4}\right) * 0.032 * N_{th}}{2.2 \times 10^{-9}}$$

$$I_{th} = 18.03 mA$$

- **Theoretical Plot Verification:**

The theoretical behavior was also plotted mathematically to validate the numerical sweep. As shown below, both the hand-calculated value and the theoretical curve align perfectly with the practical simulation, confirming a threshold current of 18.03 mA.

- **Plot the Output Power P_{out} vs. Current I (theoretical)**



Appendix (A): Source Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint
from consts import *

# define input current
I_input = 50e-3

def dydt(y,t,current):
    N, S = y
    gain = B * max(N - N0, 0)
    injection =current/(q*V)
    dndt =injection-(N/TAU)-GAMMA*C*gain*S
    dsdt = GAMMA*C*gain*S-(S/tau_ph)+(BETA_SP* N /TAU)
    return [dndt, dsdt]

t1=np.linspace(0,1e-9,2000)
t2=np.linspace(1e-9,5e-9,10000)
initial_conditions_1 =[0,0]
soln1= odeint(dydt, initial_conditions_1,t1, args=(0.0,))
initial_conditions_2 =soln1[-1,:]
soln2=odeint(dydt, initial_conditions_2,t2,args=(I_input,))
t_total= np.concatenate((t1, t2))
soln_total= np.vstack((soln1, soln2))
N_t=soln_total[:,0]
S_t=soln_total[:,1]

plt.figure(figsize=(12, 6))
plt.plot(t_total*1e9, S_t)
plt.xlabel("Time (ns)")
plt.ylabel("S (photons / m2)")
plt.title("Transient Response - ID: 1903425")
plt.show()
```

Appendix (B): Source Code

```
# diff equation input for odeint function
def dydt(y, t, current):
    N, S = y
    gain = B * max(N - N0, 0)
    injection = current/(q*V)
    dndt = injection-(N/TAU)-GAMMA*vg*gain*S
    dsdt = GAMMA*vg*gain*S-(S/tau_ph)+(BETA_SP * N / TAU)
    return [dndt, dsdt]

N_sweep = 1000
I = np.linspace(0, 60e-3, N_sweep)
P = np.zeros(N_sweep)
th = 0

for i in range(N_sweep):
    t = np.linspace(0, 15e-9, 10000)
    initial_conditions = [0, 0]
    soln = odeint(dydt, initial_conditions, t, args=(I[i],))
    N_t = soln[:, 0]
    S_t = soln[:, 1]
    P[i] = 0.5*np.log(1/R)*H*freq*vg*D*W*S_t[-1]
    if (P[i] - P[i-1]) / ((I[i] - I[i-1])) > 3000*(P[1] - P[0]) / ((I[1] - I[0])) and i > 1
and th <= 0:
        th = i

plt.figure(figsize=(12, 6))
plt.plot(I*1e3, P*1e3)
plt.xlabel("I (mA)")
plt.ylabel("P (mW)")
plt.axvline(I[th]*1e3, ls='--', lw=0.8, color='r', alpha=0.96, label='I_th')
plt.title("P Vs. I | Practical")
plt.xticks([0, 10, I[th]*1e3, 20, 30, 40, 50, 60], rotation=45)
plt.legend(loc='upper left')
plt.show()


i = np.linspace(0, 60e-3, int(1e4)) # current injection access
eta = ((np.log(1/R)*H*C*vg*tau_ph)/(2*L*q*lamda))
Ps = ((BETA_SP * tau_ph*Nth*H*C*vg*V*np.log(1/R))/(2*L*lamda*TAU))
p = np.zeros(len(i))

for k in range(0, len(i)):
    if eta*(i[k] - Ith) + Ps < Ps:
        p[k] = Ps
    else:
        p[k] = eta*(i[k]-Ith)+Ps # pout = eta(i-ith) + ps
plt.figure(figsize=(12, 6))
```

```
plt.plot(i*1e3, p*1e3, lw=2)
plt.title('P Vs. I | Theoritical')
plt.ylabel('Pout(mW)')
plt.xlabel('I(mA)')
plt.ylim([-0.1*max(p*1e3), 1.1*max(p*1e3)])
plt.axvline(Ith*1e3, ls='--', lw=0.8, color='r', alpha=0.96, label='I_th')
plt.xticks([0, 10, Ith*1e3, 20, 30, 40, 50, 60], rotation=45)
plt.legend(loc='upper left')
plt.show()
```

- **Constants file:**

```
import numpy as np

X = 4
Y = 4
L = (200+30*X)*1e-4
R = 0.3+0.02*Y
W = 2e-4
D = 0.2e-4
GAMMA = 0.3
NG = 4
ALPHA_INT = 40
B = 2.5e-16
NO = 1e18 # transparency carrier density
TAU = 2.2e-9 # spontaneous lifetime
BETA_SP = 1e-4 # spontaneous emission factor
H = 6.62607015e-34
C = 3e10

vg = C/NG # group velocity
ALPHA_M=(1/(2*L))*(np.log((1/(R*R))))
alpha_tot = ALPHA_INT + ALPHA_M # total alpha loss
V = L * D * W # volume of the diode
tau_ph = 1/(vg*alpha_tot) # photon lifetime
g_th = alpha_tot / GAMMA; # gain
q = 1.6*1e-19 # electric charge
lamda = 1.55e-4 # operating wavelength (assumed to be 1.55um)
freq = C/lamda

Nth = 1/(GAMMA * B * vg * tau_ph) + 1e18  # carrier density at threshold
Jth = Nth * q * D / TAU  # current density at threshold
Ith = Jth * L * W  # threshold current
```

- **Github Repository:**
  - https://github.com/Hazemiano/Semiconductor-Laser-Simulation